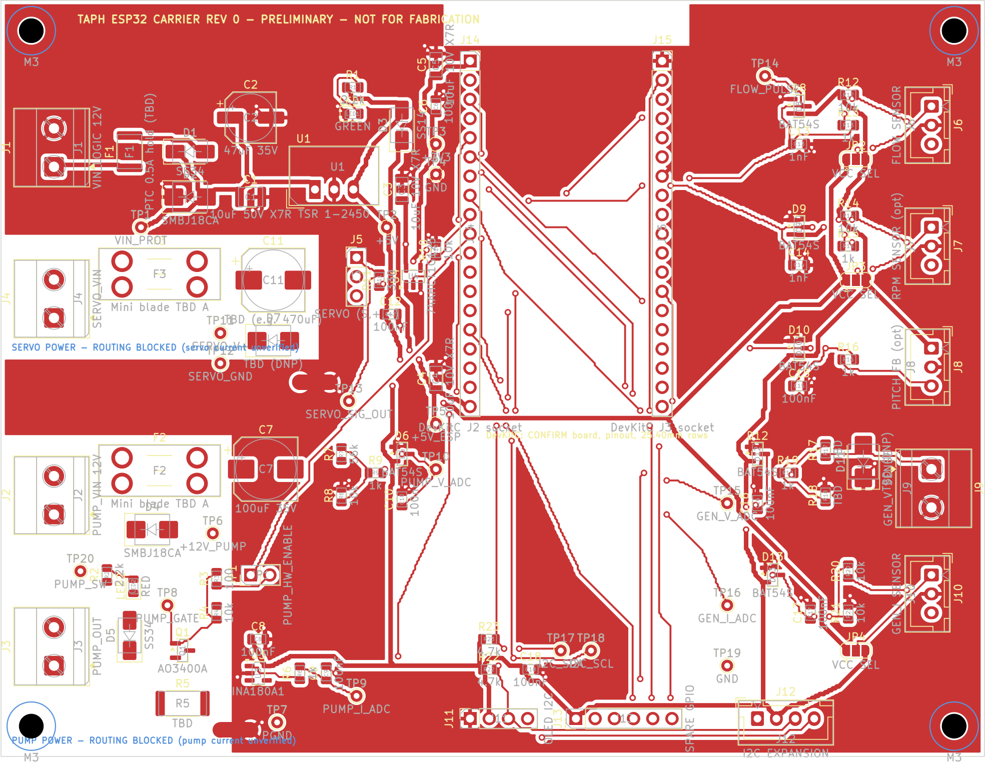
TAPH ESP32 CARRIER REV 0 – PRELIMINARY – NOT FOR FABRICATION
M3
J1
VIN_LOGIC 12V
F1
PTC 0.5A hold (TBD)
D1
SS34
D2
SMBJ18CA
C2
47uF 35V
C1
10uF 50V X7R
U1
TSR 1-2450
TP1
VIN_PROT
TP2
+5V
R1
2.2k
LED1
GREEN
D3
SS14
C5
10uF 10V X7R
C6
100nF
TP3
+3V3
TP4
GND
J14
J15
J4
SERVO_VIN
F3
Mini blade TBD A
C11
TBD (e.g. 470uF)
D7
TBD (DNP)
TP11
SERVO_V
TP12
SERVO_GND
SERVO POWER – ROUTING BLOCKED (servo current unverified)
J5
SERVO (S,+
C12
100nF
U4
74AHCT1G125
R10
10k
R11
TP13
SERVO_SIG_OUT
C4
10uF 10V X7R
TP5
DevKitC J2 socket
DevKitC: CONFIRM board, pinout, 25.40mm rows
DevKitC J3 socket
J2
PUMP_VIN 12V
F2
Mini blade TBD A
C7
100uF 35V
D4
SMBJ18CA
TP6
+12V_PUMP
TP20
PUMP_SW
R2
2.2k
LED2
RED
TP8
PUMP_GATE
R3
100
R4
10k
JP1
PUMP_HW_ENABLE
J3
PUMP_OUT
D5
SS34
Q1
AO3400A
R5
TBD
C8
100nF
U2
INA180A1
R6
1k
C9
100nF
TP9
PUMP_I_ADC
TP7
PGND
PUMP POWER – ROUTING BLOCKED (pump current unverified)
D6
+5V_ESP
R7
56k
R8
10k
R9
1k
BAT54S
TP10
PUMP_V_ADC
C10
100nF
R23
4.7k
R22
4.7k
C18
100nF
TP17
TP18
I2C_SDA
I2C_SCL
J11
OLED I2C
SPARE GPIO
J12
I2C EXPANSION
TP14
FLOW_PULSE
D8
BAT54S
C13
1nF
R12
10k
R13
1k
JP2
VCC_SEL
J6
FLOW SENSOR
D9
BAT54S
C14
1nF
R14
10k
R15
1k
JP3
VCC_SEL
J7
RPM SENSOR (opt)
D10
BAT54S
C15
100nF
R16
1k
J8
PITCH FB (opt)
D12
BAT54S
C16
100nF
R17
R18
TBD
R19
1k
D11
TBD
TP15
GEN_V_ADC
J9
GEN_VTB (DNP)
D13
BAT54S
C17
100nF
R20
10k
R21
10k
TP16
GEN_I_ADC
JP4
VCC_SEL
J10
GEN_I SENSOR
TP19
GND